TOPS Technologies

Interview Questions — DATA VISUALIZATION WITH SEABORN

Total Questions: 43

---

1. What is Seaborn, and why would you use it instead of Matplotlib for data visualization?

> **Hint:** Think about default styles and ease of creating statistical plots.

---

2. Explain the difference between figure-level and axes-level functions in Seaborn.

> **Hint:** Consider how each manages the overall plot and axes.

---

3. How do Seaborn themes like 'darkgrid', 'whitegrid', and 'ticks' affect the appearance of a plot?

> **Hint:** Focus on background, gridlines, and tick styling.

---

4. Write the code to create a simple line plot of sales over months using Seaborn.

> **Hint:** Use sns.lineplot().

---

5. What does sns.countplot() do, and when would you use it?

> **Hint:** Think about categorical data and frequency counts.

---

6. Describe the difference between sns.countplot() and sns.barplot().

> **Hint:** Consider what data each function expects and displays.

---

7. How can you change the estimator in sns.barplot() to display the sum instead of the mean?

> **Hint:** Look for the 'estimator' parameter.

---

8. Write code to create a barplot showing total sales by region, removing the confidence intervals.

> **Hint:** Check the ci parameter in sns.barplot().

---

9. What is the purpose of sns.histplot(), and what kind of data is it best suited for?

---

10. How does sns.kdeplot() differ from sns.histplot(), and what does it represent?

> **Hint:** Think about the type of distribution each plot shows.

---

11. How would you combine a histogram and a KDE plot on the same figure in Seaborn?

> **Hint:** Overlay the two plots by calling both functions on the same axes.

---

12. Write code to visualize the distribution of transaction amounts using a histogram with a KDE overlay.

---

13. What does a boxplot show, and how can it help in detecting outliers?

> **Hint:** Focus on quartiles and points outside the whiskers.

---

14. Describe the main difference between a boxplot and a violin plot.

> **Hint:** One shows distribution shape, the other focuses on quartiles.

---

15. Write code to create a boxplot of customer balances grouped by education level.

---

16. How does sns.pairplot() help in multivariate data analysis?

> **Hint:** Consider what it shows about relationships between multiple variables.

---

17. What is the purpose of the 'hue' parameter in sns.pairplot(), and how does it affect the plots?

---

18. If you want to visualize pairwise relationships among all numeric columns in a DataFrame, which Seaborn function would you use?

---

19. Explain the difference between scatter, hexbin, and KDE types in sns.jointplot().

> **Hint:** Think about how each represents point density and distribution.

---

20. Write code to create a jointplot showing the relationship between sales and profit with a regression line.

---

21. What is a correlation matrix, and how can you visualize it using a Seaborn heatmap?

> **Hint:** Think about the relationship between variables and color-coded grids.

---

22. How do you add value annotations (numbers) to each cell in a Seaborn heatmap?

---

23. Why might you use a diverging color palette for a correlation heatmap?

---

24. Write code to generate a heatmap of correlations for all numerical columns in a DataFrame, with annotations.

---

25. What is the difference between sns.relplot(), sns.scatterplot(), and sns.lineplot()?

> **Hint:** Consider figure-level vs axes-level and the types of data each is suited for.

---

26. How would you use Seaborn to plot a trend line of monthly sales by product category?

---

27. What is the purpose of regression plots like sns.lmplot() and sns.regplot()?

> **Hint:** Think about analyzing relationships and fitting lines.

---

28. How can you add Lowess smoothing to a regression plot in Seaborn?

> **Hint:** Look for a specific parameter in sns.regplot() or sns.lmplot().

---

29. Write code to create a regression plot of sales versus discount, faceted by region.

---

30. How do you set a global Seaborn style and color palette for all your plots?

> **Hint:** Check the sns.set() function.

---

31. What are some built-in Seaborn color palettes, and when would you use the 'colorblind' palette?

---

32. How does the 'context' parameter in Seaborn affect your plots, and what are some available options?

> **Hint:** Think about plot sizing for different presentation types.

---

33. Write code to create a Seaborn plot with a corporate-style theme using a custom color palette.

---

34. How can you export a Seaborn plot as a high-resolution image suitable for print?

> **Hint:** Look for the savefig() function and DPI settings.

---

35. Why would you choose SVG format when exporting a Seaborn chart for a presentation?

---

36. How can you create and reuse a template Seaborn theme for multiple plots in a report?

---

37. In a case study analyzing bank marketing data, how would you use Seaborn to identify which customer groups are more likely to subscribe to a term deposit?

> **Hint:** Think about comparing categorical and numerical features using plots.

---

38. Describe how you would perform exploratory data analysis (EDA) with Seaborn for a finance industry dataset containing both categorical and numerical columns.

---

39. If you notice that your Seaborn plots look different on another computer or in a Jupyter notebook, what settings could you check or adjust to ensure consistency?

---

40. How could you use ChatGPT or another AI assistant to help you write or debug Seaborn visualization code, and what steps would you take to verify the AI’s suggestions?

> **Hint:** Consider running the code, checking documentation, and reviewing plot output.

---

41. What are some ways AI image generation tools could assist in creating enhanced visuals for your data analysis reports, and what limitations should you be aware of?

---

42. Explain how you would combine a heatmap and a pairplot in a single analysis to provide deeper insights about relationships in your data.

> **Hint:** Think about what each plot type reveals and how they complement each other.

---

43. Describe a scenario where you would use both a boxplot and a violin plot to analyze a dataset, and what extra information the violin plot provides.
